LATENCY-AWARE CLOUD TASK PLACEMENT

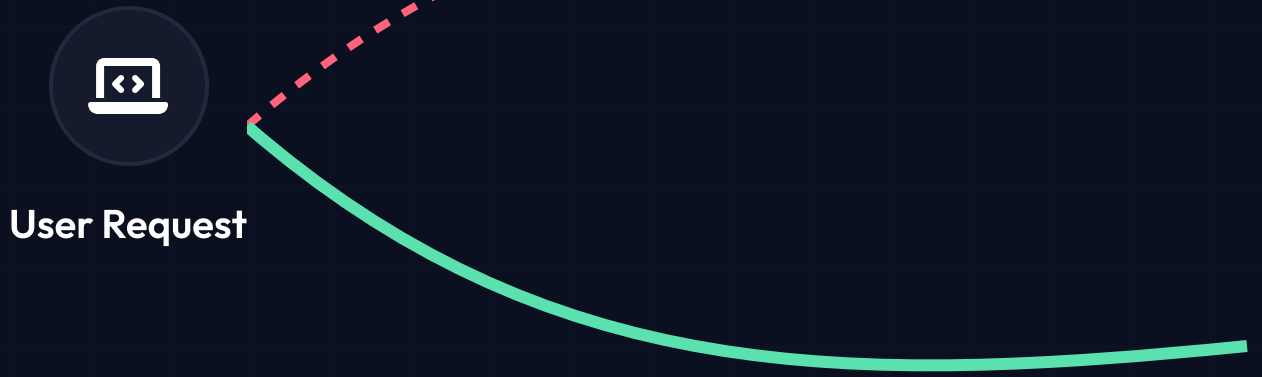
LIGHTPATH

Smarter Placement. **Shorter Paths.** Faster Responses.

THE CHALLENGE

The Problem with Cloud Placement

Assigning tasks to servers is not as simple as choosing any available machine. Poor placement decisions dramatically increase response times and network overhead.



Server A Inefficient

Farther away or connected through a higher-latency route.

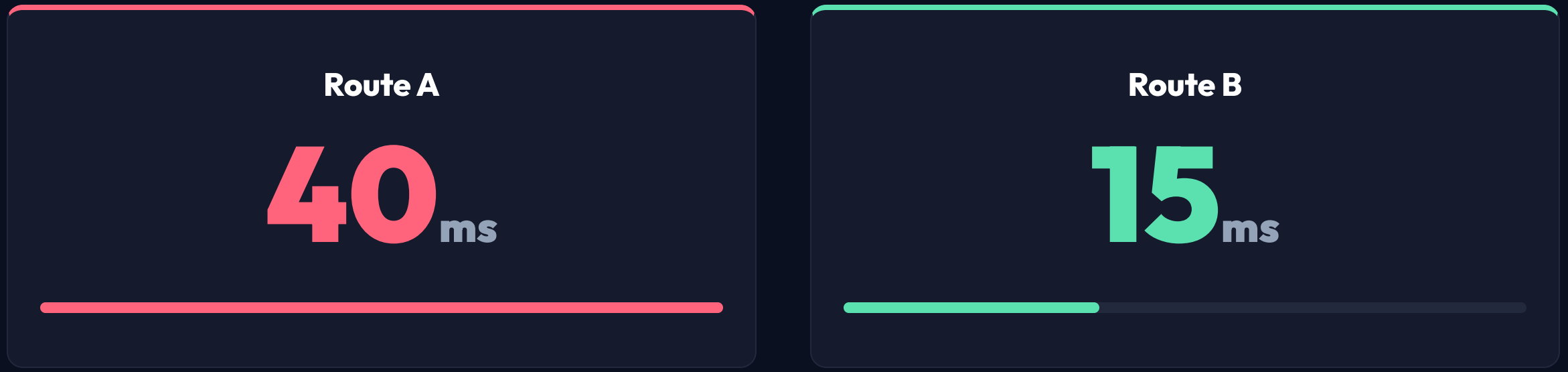
Server B Optimal

A potentially better placement with a shorter effective communication delay.

CORE CONCEPT

What Is Latency?

The time delay involved in transmitting data or obtaining a response across a network.



"Would you choose the nearest server geographically, or the route with the lowest total latency?"

** Clearly labeled illustrative hypothetical examples, not measured results from the project.*

Understanding Task Placement

An incoming computational task must be matched with a suitable computing resource.



Good placement must consider **communication delay** as well as **resource availability**.

SYSTEM ARCHITECTURE

How Lightpath Works Conceptually

01

Map Infrastructure

Represent the infrastructure as a network of connected nodes.

02

Analyze Task

Identify the computational task and its specific computing requirements.

03

Evaluate Paths

Calculate possible destinations and their corresponding communication paths.

04

Compare Options

Weigh feasible placement options against each other based on metrics.

05

Select & Execute

Select destination based on constraints, execute placement, and evaluate performance.

** Note: This is a conceptual workflow.*

THE ENGINE

Algorithms and Optimization

Graph Representation

Servers and network connections modeled as a weighted graph.

Shortest-Path Algorithms

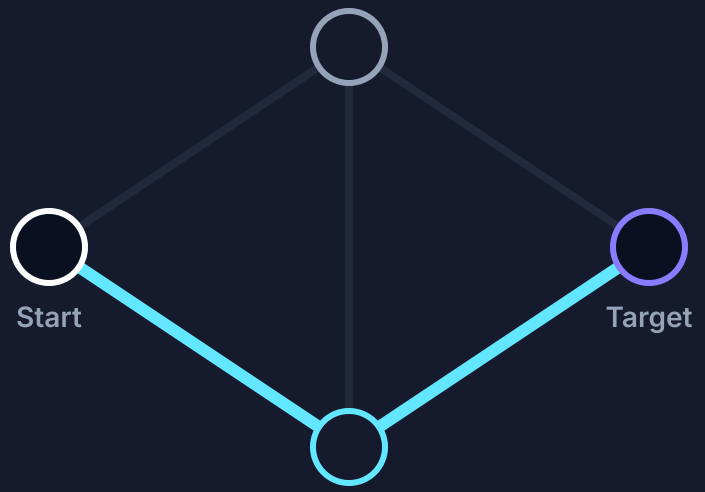
Finding a low-cost, low-latency route through a network.

Resource Allocation

Ensuring a chosen server has sufficient processing capacity.

Optimization

Balancing strict latency requirements with other system constraints.



INTERACTIVE SCENARIO

Route Selection Comparison

Inefficient Placement

- Task assigned to a less suitable destination.
- Longer or higher-latency communication route.
- Potentially slower response.



Improved Placement

- Task assigned to a feasible, better-suited destination.
- Lower-latency communication route in the illustrative scenario.
- Potentially faster response.



ANALYTICS

Performance Evaluation

Evaluating a task-placement system requires monitoring key metrics.

METRIC 1

Communication Latency

METRIC 2

Task Completion Time

METRIC 3

Server Utilization

METRIC 4

Resource Feasibility

Important Note

The charts above contain illustrative/hypothetical data and are **not measured project results**. A valid comparison requires consistent workloads, network conditions, and evaluation methods.

LOOKING FORWARD

Applications and Future Scope

Potential Applications

- Cloud-hosted web applications
- Online gaming & interactive services
- Video streaming
- Distributed & Edge computing

Proposed Future Improvements

(Distinguished from existing implementation features)

Dynamic Workload Adaptation

Adjusting routing algorithms based on real-time network fluctuations.

Energy-Aware Placement

Factoring power consumption and green energy availability into node selection.

Multi-Objective Optimization

Simultaneously balancing latency, throughput, cost, and availability.

Key Takeaways

1. Where a task runs affects its communication delay.
2. Network paths and server capacity both matter.
3. Latency-aware placement can help improve application responsiveness when the chosen placement is feasible and the optimization is effective.

THE RIGHT TASK. THE RIGHT SERVER. THE RIGHT PATH.

Questions & Answers